

3. Pandas

Pandas je još jedna u nizu gradivnih bibilioteka za analizu podataka korišćenjem Python-a. U pitanju je moćna, brza i veoma fleksibilna biblioteka koja omogućava manipulaciju sa brojnim strukturama - jedna će biti od posebnog značaja za nas, Pandas Datafame. Dataframe je zapravo dvodimenziona struktura, poput matrice, u kojoj su kolone neki atributi posmatranih podataka a redovi samo opservacije, uzorci, pojavljivanja našeg predmetnog istraživanja. Recimo da istražujemo neko tržište i vršimo analizu podataka o kupcima. Jedan dataframe bi stoga izgledao tako što bi svaki njegov red bio poseban kupac dok bi kolone predstavljale neke atribute datog kupca - količina novca koji troši na sedmičnoj bazi, broj godina, kategorija proizvoda koju najviše kupuje i slično. Još jedna od ogromih pogodnosti Pandas paketa je fleksibilnost po pitanju reprezentacije podataka - Pandas ih učita u tabelarnoj formi iz brojnih formata - .csv, .tsv, .xlsx i slično.

Sekcija 1 Pandas - osnovne komande nad dataframe strukturom

Počnimo naredbom za učitavanje Pandas paketa:

In [1]:
```
import pandas as pd
```

Ovo je standardna komanda za učitavanje pandas-a. `pd` je skraćenica kojom ćemo lakše pozivati pandas kasnije u kodu. Fokusirajmo se odmah na učitavanje jednog `dataframe-a` iz `.csv` formata.

In [61]:
```
df = pd.read_csv("Salaries.csv")
```

U promenljivoj df sada imamo dataframe - tabelarni prikaz podataka iz `.csv` formata. Ispitajmo izgled dataframe-a sa funkcijom head, koja štampa prvih nekoliko redova dataframe-a.

In [3]:
```
df.head()
```

Out[3]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	Tota
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.00	400184.25	NaN	56759
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	53890
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	NaN	33527
3	4	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC	77916.00	56120.71	198306.90	NaN	33234
4	5	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	134401.60	9737.00	182234.59	NaN	32637

Vidimo znatan broj podataka. Očigledno da su u pitanju podaci o platama, beneficijama, bonusima i prekovremenom radu - ukratko o ukupnom plaćanju radnika na platnom spisku neke kompanije. Istražićemo Pandas uz pomoć ovog dataframe-a. Možemo štampati nekoliko poslednjih redova uz pomoć funkcije `tail` ili specificirati veći broj redova za štampu u head ili tail funkciji.

In [4]:
```
df.tail()
```

Out[4]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	To
148649	148650	Roy I Tillery	Custodian	0.0	0.0	0.00	0.0	
148650	148651	Not provided	Not provided	NaN	NaN	NaN	NaN	
148651	148652	Not provided	Not provided	NaN	NaN	NaN	NaN	
148652	148653	Not provided	Not provided	NaN	NaN	NaN	NaN	
148653	148654	Joe Lopez	Counselor, Log Cabin Ranch	0.0	0.0	-618.13	0.0	

In [5]:
```
df.head(10)
```

Out[5]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	Tota
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.00	400184.25	NaN	5675
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	5389
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	NaN	3352
3	4	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC	77916.00	56120.71	198306.90	NaN	3323
4	5	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	134401.60	9737.00	182234.59	NaN	3263
5	6	DAVID SULLIVAN	ASSISTANT DEPUTY CHIEF II	118602.00	8601.00	189082.74	NaN	3162
6	7	ALSON LEE	BATTALION CHIEF, (FIRE DEPARTMENT)	92492.01	89062.90	134426.14	NaN	3159
7	8	DAVID KUSHNER	DEPUTY DIRECTOR OF INVESTMENTS	256576.96	0.00	51322.50	NaN	3078
8	9	MICHAEL MORRIS	BATTALION CHIEF, (FIRE DEPARTMENT)	176932.64	86362.68	40132.23	NaN	3034
9	10	JOANNE HAYES-WHITE	CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	285262.00	0.00	17115.73	NaN	3023

Recimo da kao kod NumPy biblioteke želimo da proverimo broj dimenzija jednog dataframe-a. Funkcija `shape` vraća tuple u kome prva cifra pokazuje broj redova, tj. opservacija a druga cifra broj kolona tj. atributa.

In [7]:
```
opservacije,atributi = df.shape
print(f'Naš dataframe ima {opservacije} opservacija i {atributi} atributa.')
```

```
Naš dataframe ima 148654 opservacija i 13 atributa.
```

Uz pomoć funkcije `info` možemo da imamo informacije o samim kolonama - tip podatka, broj vrednosti koje nisu nedostajuće itd.

In [8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148654 entries, 0 to 148653
Data columns (total 13 columns):
 #   Column            Non-Null Count   Dtype  
---  ------            --------------   -----  
 0   Id                148654 non-null  int64  
 1   EmployeeName      148654 non-null  object 
 2   JobTitle          148654 non-null  object 
 3   BasePay           148045 non-null  float64
 4   OvertimePay       148650 non-null  float64
 5   OtherPay          148650 non-null  float64
 6   Benefits          112491 non-null  float64
 7   TotalPay          148654 non-null  float64
 8   TotalPayBenefits  148654 non-null  float64
 9   Year              148654 non-null  int64  
 10  Notes             0 non-null       float64
 11  Agency            148654 non-null  object 
 12  Status            0 non-null       float64
dtypes: float64(8), int64(2), object(3)
memory usage: 14.7+ MB
```

Vidimo do sada neviđen tip podatka - `object`. U kontekstu dataframe-a ovo predstavlja kategorički atribut.

Možemo odvojeno vratiti imena kolona dataframe-a ili tipove vrednosti:

In [9]:
```
df.columns
```

Out[9]:
```
Index(['Id', 'EmployeeName', 'JobTitle', 'BasePay', 'OvertimePay', 'OtherPay',
       'Benefits', 'TotalPay', 'TotalPayBenefits', 'Year', 'Notes', 'Agency',
       'Status'],
      dtype='object')
```

In [10]:
```
df.dtypes
```

Out[10]:
```
Id                    int64
EmployeeName         object
JobTitle             object
BasePay             float64
OvertimePay         float64
OtherPay            float64
Benefits            float64
TotalPay            float64
TotalPayBenefits    float64
Year                  int64
Notes               float64
Agency               object
Status              float64
dtype: object
```

Od velike važnosti nam je funkcija `select_dtypes` - pruža nam mogućnost da izdvojimo manji dataframe iz prvobitnog po kriterijumu tipa podatka. Recimo da želimo da izdvojimo numeričke podatke.

In [13]:
```
num_df = df.select_dtypes(include = ['int64','float64'])
#num_df = df.select_dtypes(exclude = ['object']) alternativno
```

In [14]:
```
num_df.head()
```

Out[14]:

	Id	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits	Year	Not
0	1	167411.18	0.00	400184.25	NaN	567595.43	567595.43	2011	N
1	2	155966.02	245131.88	137811.38	NaN	538909.28	538909.28	2011	N
2	3	212739.13	106088.18	16452.60	NaN	335279.91	335279.91	2011	N
3	4	77916.00	56120.71	198306.90	NaN	332343.61	332343.61	2011	N
4	5	134401.60	9737.00	182234.59	NaN	326373.19	326373.19	2011	N

Vidimo da smo uspešno selektovali samo numerčke podatke iz dataframe-a. Respektivno, ako koristimo `exclude` umesto `include` kod prvog parametra funkcije možemo dobiti one tipove podataka koji nisu navedeni u parametru funkcije. Još jedna zgodna funkcija je funkcija describe koja nam praktično daje izveštaj o deskriptivnim statistikama neke numeričke kolone. Kako smo mi izdvojili manji dataframe `num_df` koji se sastoji samo od numeričkih podataka, možemo primeniti funkciju na njega.

In [15]:
```
num_df.describe()
```

Out[15]:

	Id	BasePay	OvertimePay	OtherPay	Benefits	To
count	148654.000000	148045.000000	148650.000000	148650.000000	112491.000000	148654.
mean	74327.500000	66325.448840	5066.059886	3648.767297	25007.893151	74768.
std	42912.857795	42764.635495	11454.380559	8056.601866	15402.215858	50517.
min	1.000000	-166.010000	-0.010000	-7058.590000	-33.890000	-618.
25%	37164.250000	33588.200000	0.000000	0.000000	11535.395000	36168.
50%	74327.500000	65007.450000	0.000000	811.270000	28628.620000	71426.
75%	111490.750000	94691.050000	4658.175000	4236.065000	35566.855000	105839.
max	148654.000000	319275.010000	245131.880000	400184.250000	96570.660000	567595.

Sekcija 2 Pandas - selekcija i "slajsovanje" dataframe strukture

Određene kolone iz dataframe-a možemo selektovati pomoću njihovog naziva:

In [16]:
```
df['BasePay']
```

Out[16]:
```
0         167411.18
1         155966.02
2         212739.13
3          77916.00
4         134401.60
             ...
148649         0.00
148650          NaN
148651          NaN
148652          NaN
148653         0.00
Name: BasePay, Length: 148654, dtype: float64
```

Možemo uzeti i listu kolona:

In [17]:
```
df[['BasePay','OvertimePay','OtherPay']]
```

Out[17]:

	BasePay	OvertimePay	OtherPay
0	167411.18	0.00	400184.25
1	155966.02	245131.88	137811.38
2	212739.13	106088.18	16452.60
3	77916.00	56120.71	198306.90
4	134401.60	9737.00	182234.59
...	...	...	...
148649	0.00	0.00	0.00
148650	NaN	NaN	NaN
148651	NaN	NaN	NaN
148652	NaN	NaN	NaN
148653	0.00	0.00	-618.13

148654 rows × 3 columns

Pokažimo sada kako možemo izbaciti određenu kolonu preko funkcije `drop`. Rezultat se mora upisati nazad u prom. df. ili se mora specificirati `inplace` u parametrima drop funkcije - `inplace = True`

In [62]:
```
df = df.drop('TotalPay', axis = 1) #axis označava osu po kojoj tražimo naziv totalp
#alternativa : df.drop('TotalPay', axis = 1, inplace=True)
```

In [23]:
```
df.head(3)
```

Out[23]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	Totall
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.00	400184.25	NaN	
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	NaN	

Vidimo da smo ostali bez date promenljive. Želimo da je kreiramo ponovo. Dodajemo novu kolonu vrlo lako. Prosleđujemo neki naziv i sabiramo basePay, OvertimePay i OtherPay.

In [63]:
```
df['TotalPayNew'] = df['BasePay'] + df['OvertimePay'] + df['OtherPay']
df.head(3)
```

Out[63]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	Totall
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.00	400184.25	NaN	
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	NaN	

Na kraj dataframe-a dodata je nova kolona.
Redove možemo birati po nekom logičkom uslovu. Recimo da želimo samo instance čiji je `BasePay` veći od dvesta hiljada.

In [25]:
```
df[df['BasePay']>200000]
```

Out[25]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Be
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	
7	8	DAVID KUSHNER	DEPUTY DIRECTOR OF INVESTMENTS	256576.96	0.00	51322.50	
9	10	JOANNE HAYES-WHITE	CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	285262.00	0.00	17115.73	
12	13	EDWARD HARRINGTON	EXECUTIVE CONTRACT EMPLOYEE	294580.02	0.00	0.00	
13	14	JOHN MARTIN	DEPARTMENT HEAD V	271329.03	0.00	21342.59	
...	...	...	...	...	...	...	
110953	110954	Ronald P Flynn	Assistant Chief Attorney 1	201296.01	0.00	1250.00	571
110954	110955	Sheryl L Bregman	Assistant Chief Attorney 1	201296.00	0.00	1250.00	571
110976	110977	Susan T Obata	Supervising Physician Spec	202445.22	0.00	0.00	556
110977	110978	John O Funghi	Engineer/Architect Principal	201181.01	0.00	0.00	568
111107	111108	Mayra T Vivas	Senior Physician Specialist	205036.00	0.00	1260.00	452

509 rows × 13 columns

Vidimo da smo izdvojili neki manji podskup u podacima - svega 509 redova.
Možemo zadavati više logičkih uslova.

In [28]:
```
df[(df['BasePay']>200000) & (df['JobTitle'].str.contains('Assistant'))]
```

Out[28]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	T
36179	36180	Michael Morris	Assistant Deputy Chief 2	223371.00	7415.52	63426.03	56202.98	
36190	36191	Thomas Harvey	Assistant Deputy Chief 2	226475.05	0.00	54967.53	55722.16	
36233	36234	Jose Velo	Assistant Deputy Chief 2	226475.02	0.00	16877.44	58449.76	
36325	36326	Chris Vein	Special Assistant 22	203704.37	0.00	3486.00	68285.81	
72975	72976	Kenneth A Lombardi	Assistant Deputy Chief 2	239247.00	0.00	19855.26	68856.99	
72982	72983	Jose L Velo	Assistant Deputy Chief 2	239247.05	0.00	14354.82	70323.94	
72986	72987	Kyle J Merkins	Assistant Deputy Chief 2	239247.04	0.00	14354.82	67951.69	
73127	73128	Jennifer E Matz	Special Assistant 22	216876.76	0.00	0.00	63846.54	
73216	73217	David A Pfeifer	Assistant Chief Attorney 2	213587.50	0.00	1250.00	56230.97	
73223	73224	Noreen M Ambrose	Assistant Chief Attorney 2	213587.55	0.00	1249.50	55829.30	
73224	73225	Robert S Maerz	Assistant Chief Attorney 2	213587.52	0.00	1249.50	55829.30	
73225	73226	Matthew J Rothschild	Assistant Chief Attorney 2	213587.51	0.00	1249.50	55829.30	
73229	73230	Sharon Woo	Assistant Chief Attorney 2	213587.51	0.00	1250.00	55589.40	

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	T
73272	73273	Kimiko C Burton	Assistant Chief Attorney 2	211929.88	0.00	1249.50	55024.42	
73323	73324	Tiffany J Bohee	Special Assistant 21	201952.51	0.00	0.00	63103.46	
73404	73405	Julie Van Nostern	Assistant Chief Attorney 1	203425.10	0.00	1249.50	54158.47	
73407	73408	June D Cravet	Assistant Chief Attorney 1	203425.09	0.00	1250.00	53882.70	
73408	73409	Kate Herrman Stacy	Assistant Chief Attorney 1	203425.09	0.00	1249.50	53882.61	
73409	73410	Donald P Margolis	Assistant Chief Attorney 1	203425.06	0.00	1249.50	53882.61	
73411	73412	Jerry P Coleman	Assistant Chief Attorney 1	203425.01	0.00	1250.00	53825.12	
73413	73414	Theresa L Mueller	Assistant Chief Attorney 1	203425.11	0.00	1249.50	53642.64	
73414	73415	James Emery	Assistant Chief Attorney 1	203425.10	0.00	1249.50	53642.63	
73415	73416	Julia M C Friedlander	Assistant Chief Attorney 1	203425.06	0.00	1249.50	53642.63	
73416	73417	Robert A Bryan	Assistant Chief Attorney 1	203425.06	0.00	1249.50	53642.63	
73417	73418	Aleeta Van Runkle	Assistant Chief	203425.06	0.00	1249.50	53642.63	

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	T
			Attorney 1					
73418	73419	Ronald P Flynn	Assistant Chief Attorney 1	203425.01	0.00	1249.50	53642.63	
73419	73420	Owen J Clements	Assistant Chief Attorney 1	203425.01	0.00	1249.50	53642.63	
73420	73421	Wayne K Snodgrass	Assistant Chief Attorney 1	203425.08	0.00	1249.50	53546.43	
73422	73423	Paul D Henderson	Assistant Chief Attorney 1	203425.02	0.00	1250.00	53370.78	
73454	73455	David M Merin	Assistant Chief Attorney 1	201283.02	0.00	1250.00	53107.40	
110576	110577	Kenneth A Lombardi	Assistant Deputy Chief 2	230386.00	0.00	19119.88	75049.76	
110768	110769	Robert S Maerz	Assistant Chief Attorney 2	211360.01	0.00	1918.00	59421.80	
110773	110774	Sharon Woo	Assistant Chief Attorney 2	211360.04	0.00	1312.50	59295.54	
110776	110777	Kimiko C Burton	Assistant Chief Attorney 2	211360.06	0.00	1250.00	59275.24	
110777	110778	Noreen M Ambrose	Assistant Chief Attorney 2	211360.06	0.00	1250.00	59275.24	
110778	110779	Matthew J Rothschild	Assistant Chief Attorney 2	211360.04	0.00	1250.00	59275.24	

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	T
110859	110860	Jennifer E Matz	Special Assistant 22	200308.31	0.00	0.00	64380.91	
110892	110893	James Emery	Assistant Chief Attorney 1	201296.10	0.00	3126.00	57537.24	
110893	110894	Aleeta Van Runkle	Assistant Chief Attorney 1	201296.07	0.00	3126.00	57537.22	
110894	110895	Robert A Bryan	Assistant Chief Attorney 1	201296.06	0.00	3126.00	57537.21	
110895	110896	Julia M C Friedlander	Assistant Chief Attorney 1	201296.04	0.00	3126.00	57537.21	
110896	110897	Wayne K Snodgrass	Assistant Chief Attorney 1	201296.04	0.00	3126.00	57537.19	
110897	110898	Theresa L Mueller	Assistant Chief Attorney 1	201296.01	0.00	3126.00	57537.17	
110898	110899	Julie Van Nostern	Assistant Chief Attorney 1	201296.00	0.00	3126.00	57537.17	
110899	110900	Owen J Clements	Assistant Chief Attorney 1	201296.00	0.00	3126.00	57537.17	
110941	110942	David M Merin	Assistant Chief Attorney 1	201296.01	0.00	1312.50	57139.33	
110942	110943	June D Cravet	Assistant Chief Attorney 1	201296.00	0.00	1312.50	57139.33	
110943	110944	Paul D Henderson	Assistant Chief	201296.00	0.00	1312.50	57139.33	

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	T
			Attorney 1					
110944	110945	Eileen M Malley	Assistant Chief Attorney 1	201343.81	0.00	1250.00	57138.91	
110949	110950	Cheryl S Adams	Assistant Chief Attorney 1	201296.02	0.00	1250.00	57125.64	
110950	110951	Jonathan Givner	Assistant Chief Attorney 1	201296.02	0.00	1250.00	57125.63	
110951	110952	Kate Herrman Stacy	Assistant Chief Attorney 1	201296.02	0.00	1250.00	57125.62	
110952	110953	Yvonne R Mere	Assistant Chief Attorney 1	201296.02	0.00	1250.00	57125.62	
110953	110954	Ronald P Flynn	Assistant Chief Attorney 1	201296.01	0.00	1250.00	57125.63	
110954	110955	Sheryl L Bregman	Assistant Chief Attorney 1	201296.00	0.00	1250.00	57125.62	

Možemo zadati logički uslov i izabrati samo jedan deo kolona:

In [29]:
```
df[(df['BasePay']>200000) & (df['JobTitle'].str.contains('Assistant'))][['EmployeeN
```

Out[29]:

	EmployeeName	JobTitle
36179	Michael Morris	Assistant Deputy Chief 2
36190	Thomas Harvey	Assistant Deputy Chief 2
36233	Jose Velo	Assistant Deputy Chief 2
36325	Chris Vein	Special Assistant 22
72975	Kenneth A Lombardi	Assistant Deputy Chief 2
72982	Jose L Velo	Assistant Deputy Chief 2
72986	Kyle J Merkins	Assistant Deputy Chief 2
73127	Jennifer E Matz	Special Assistant 22
73216	David A Pfeifer	Assistant Chief Attorney 2
73223	Noreen M Ambrose	Assistant Chief Attorney 2
73224	Robert S Maerz	Assistant Chief Attorney 2
73225	Matthew J Rothschild	Assistant Chief Attorney 2
73229	Sharon Woo	Assistant Chief Attorney 2
73272	Kimiko C Burton	Assistant Chief Attorney 2
73323	Tiffany J Bohee	Special Assistant 21
73404	Julie Van Nostern	Assistant Chief Attorney 1
73407	June D Cravet	Assistant Chief Attorney 1
73408	Kate Herrman Stacy	Assistant Chief Attorney 1
73409	Donald P Margolis	Assistant Chief Attorney 1
73411	Jerry P Coleman	Assistant Chief Attorney 1
73413	Theresa L Mueller	Assistant Chief Attorney 1
73414	James Emery	Assistant Chief Attorney 1
73415	Julia M C Friedlander	Assistant Chief Attorney 1
73416	Robert A Bryan	Assistant Chief Attorney 1
73417	Aleeta Van Runkle	Assistant Chief Attorney 1
73418	Ronald P Flynn	Assistant Chief Attorney 1
73419	Owen J Clements	Assistant Chief Attorney 1
73420	Wayne K Snodgrass	Assistant Chief Attorney 1
73422	Paul D Henderson	Assistant Chief Attorney 1
73454	David M Merin	Assistant Chief Attorney 1

	EmployeeName	JobTitle
110576	Kenneth A Lombardi	Assistant Deputy Chief 2
110768	Robert S Maerz	Assistant Chief Attorney 2
110773	Sharon Woo	Assistant Chief Attorney 2
110776	Kimiko C Burton	Assistant Chief Attorney 2
110777	Noreen M Ambrose	Assistant Chief Attorney 2
110778	Matthew J Rothschild	Assistant Chief Attorney 2
110859	Jennifer E Matz	Special Assistant 22
110892	James Emery	Assistant Chief Attorney 1
110893	Aleeta Van Runkle	Assistant Chief Attorney 1
110894	Robert A Bryan	Assistant Chief Attorney 1
110895	Julia M C Friedlander	Assistant Chief Attorney 1
110896	Wayne K Snodgrass	Assistant Chief Attorney 1
110897	Theresa L Mueller	Assistant Chief Attorney 1
110898	Julie Van Nostern	Assistant Chief Attorney 1
110899	Owen J Clements	Assistant Chief Attorney 1
110941	David M Merin	Assistant Chief Attorney 1
110942	June D Cravet	Assistant Chief Attorney 1
110943	Paul D Henderson	Assistant Chief Attorney 1
110944	Eileen M Malley	Assistant Chief Attorney 1
110949	Cheryl S Adams	Assistant Chief Attorney 1
110950	Jonathan Givner	Assistant Chief Attorney 1
110951	Kate Herrman Stacy	Assistant Chief Attorney 1
110952	Yvonne R Mere	Assistant Chief Attorney 1
110953	Ronald P Flynn	Assistant Chief Attorney 1
110954	Sheryl L Bregman	Assistant Chief Attorney 1

Još jedan značajan način za selekciju određenih redova ili kolona su `loc` i `iloc` funkcije. `loc` funkcija koristi nazive kolona i/ili redova dok `iloc` koristi poziciju, tj. redosled. Recimo da koristimo `loc` funkciju da izaberemo prvih 20 redova i kolone EmployeeName i JobTitle isključivo:

In [30]:
```
df.loc[0:20,['EmployeeName','JobTitle']]
```

Out[30]:

	EmployeeName	JobTitle
0	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY
1	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)
2	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)
3	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC
4	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT,(FIRE DEPARTMENT)
5	DAVID SULLIVAN	ASSISTANT DEPUTY CHIEF II
6	ALSON LEE	BATTALION CHIEF, (FIRE DEPARTMENT)
7	DAVID KUSHNER	DEPUTY DIRECTOR OF INVESTMENTS
8	MICHAEL MORRIS	BATTALION CHIEF, (FIRE DEPARTMENT)
9	JOANNE HAYES-WHITE	CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)
10	ARTHUR KENNEY	ASSISTANT CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)
11	PATRICIA JACKSON	CAPTAIN III (POLICE DEPARTMENT)
12	EDWARD HARRINGTON	EXECUTIVE CONTRACT EMPLOYEE
13	JOHN MARTIN	DEPARTMENT HEAD V
14	DAVID FRANKLIN	BATTALION CHIEF, (FIRE DEPARTMENT)
15	RICHARD CORRIEA	COMMANDER III, (POLICE DEPARTMENT)
16	AMY HART	DEPARTMENT HEAD V
17	SEBASTIAN WONG	CAPTAIN, EMERGENCYCY MEDICAL SERVICES
18	MARTY ROSS	BATTALION CHIEF, (FIRE DEPARTMENT)
19	ELLEN MOFFATT	ASSISTANT MEDICAL EXAMINER
20	VENUS AZAR	ASSISTANT MEDICAL EXAMINER

Ukoliko nekim slučajem ne znamo nazive kolona ali znamo njihove pozicije - koristimo `iloc`.

In [31]:
```
df.iloc[[0,57,534,666],[1,2]]
```

Out[31]:

	EmployeeName	JobTitle
0	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY
57	MARK KEARNEY	ASSISTANT DEPUTY CHIEF II
534	ROSE QUINONES	PHYSICIAN ASSISTANT
666	SEAN CONNOLLY	HEAD ATTORNEY, CIVIL AND CRIMINAL

Sekcija 3 Pandas - nedostajuće vrednosti

Ispitajmo naš dataframe još jednom:

In [32]:
```
df.head()
```

Out[32]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	Totall
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18	0.00	400184.25	NaN	
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02	245131.88	137811.38	NaN	
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13	106088.18	16452.60	NaN	
3	4	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC	77916.00	56120.71	198306.90	NaN	
4	5	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	134401.60	9737.00	182234.59	NaN	

Vidimo da se u atributima `Notes` i `Status` javljaju do sada nepozate NaN vrednosti. NaN (Not a Number) su nedostajuće ili nepoznate vrednosti. Kako nam one nisu od važnosti među podacima koristimo različite strategije da ih uklonimo. Funkcijom `isna` možemo da proverimo tačnu strukturu NaN vrednosti na celom dataframe-u.

In [34]:
```
df.isna().sum()
```

Out[34]:
```
Id                      0
EmployeeName            0
JobTitle                0
BasePay               609
OvertimePay             4
OtherPay                4
Benefits            36163
TotalPayBenefits        0
Year                    0
Notes              148654
Agency                  0
Status             148654
TotalPayNew           609
dtype: int64
```

Vidimo da postoje nedostajuće vrednosti u kategoričkim podacima - Status i Notes i u numeričkim podacima Benefits, BasePay, OvertimePay ,Other Pay i TotalPayNew. Obradimo prvo numeričke podatke.

Kako efektivno možemo zameniti ovakve podatke? Prirodno, srednjom vrednošću date kolone.

In [64]:
```
df['BasePay'].fillna(value=df['BasePay'].mean(),inplace = True)
df['TotalPayNew'].fillna(value=df['TotalPayNew'].mean(), inplace = True)
df['Benefits'].fillna(value=df['Benefits'].mean(), inplace = True)
```

In [68]:
```
df['OvertimePay'].fillna(value=df['OvertimePay'].mean(), inplace = True)
df['OtherPay'].fillna(value=df['OtherPay'].mean(), inplace = True)
```

A sada se fokusirajmo na kategoričke podatke. Kako atributi `Notes` i `Status` imaju previše nedostajućih vrednosti odlučujemo da ih ne menjamo jer time značajno remetimo strukturu tih podataka. Umesto toga ćemo samo izbaciti date kolone.

In [65]:
```
df.drop('Notes',axis = 1, inplace = True)
df.drop('Status', axis = 1, inplace = True)
```

In [70]:
```
print(f'Broj nedostajućih vrednosti posle zamene: {df.isna().sum().sum()}.')
```

```
Broj nedostajućih vrednosti posle zamene: 0.
```

Sekcija 4 Pandas - korisne funkcije nad dataframe-om

Jedna od veoma značajnih stvari za koju Pandas poseduje implementaciju je nalaženje jedinstvenih vrednosti neke varijable i raspole iste, pokažimo na primeru:

In [71]:
```
df['Year'].unique()
```

Out[71]:
```
array([2011, 2012, 2013, 2014], dtype=int64)
```

Uviđamo značajan podatak - sve naše instance su zarade radnika iz godine 2011, 2012, 2013 ili 2014. Šta ako želimo da vidimo koliko je ovih instanci u kojoj godini?

In [72]:
```
df['Year'].value_counts()
```

Out[72]:
```
Year
2014    38123
2013    37606
2012    36766
2011    36159
Name: count, dtype: int64
```

Možemo sortirati dataframe po jednoj koloni:

In [73]:
```
df.sort_values(by='Benefits') #inplace false po default-u
```

Out[73]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Bene
110530	110531	David P Kucia	Police Officer 3	66325.44884	0.00	0.00	-33
144099	144100	Leticia Villacampa	Secretary 2	0.00000	0.00	9991.77	-13
110529	110530	Mark E Laherty	Police Officer 3	66325.44884	0.00	0.00	-8
110528	110529	Timothy E Gibson	Police Officer 3	66325.44884	0.00	0.00	-2
90855	90856	Geraldine P Hernandez	Special Nurse	81549.59000	9796.90	13595.81	0
...	...	...	...	...	...	...	
110532	110533	Amy P Hart	Asst Med Examiner	318835.49000	10712.95	60563.54	89540
110537	110538	John L Martin	Dept Head V	311298.55000	0.00	0.00	89772
110535	110536	Joanne M Hayes-White	Chief, Fire Department	302068.00000	0.00	24165.44	91201
110534	110535	Gregory P Suhr	Chief of Police	307450.04000	0.00	19266.72	91302
110533	110534	William J Coaker Jr.	Chief Investment Officer	257340.00000	0.00	82313.70	96570

148654 rows × 11 columns

Možemo nalaziti nan i null vrednosti preko sledećih funkcija:

In [ ]:
```
df.isnull()
df.isna()
```

Možemo izbaciti redove koji makar u jednoj koloni imaju nan vrednosti sledećom funkcijom:

In [74]:
```
df.dropna() #napomena: funkciji se mogu proslediti dodatni parametri koji racunaju
#o broju ned. vrednosti itd.
```

Out[74]:

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherP
0	1	NATHANIEL FORD	GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY	167411.18000	0.000000	400184.2500
1	2	GARY JIMENEZ	CAPTAIN III (POLICE DEPARTMENT)	155966.02000	245131.880000	137811.3800
2	3	ALBERT PARDINI	CAPTAIN III (POLICE DEPARTMENT)	212739.13000	106088.180000	16452.6000
3	4	CHRISTOPHER CHONG	WIRE ROPE CABLE MAINTENANCE MECHANIC	77916.00000	56120.710000	198306.9000
4	5	PATRICK GARDNER	DEPUTY CHIEF OF DEPARTMENT, (FIRE DEPARTMENT)	134401.60000	9737.000000	182234.5900
...	...	...	...	...	...	
148649	148650	Roy I Tillery	Custodian	0.00000	0.000000	0.0000
148650	148651	Not provided	Not provided	66325.44884	5066.059886	3648.7672
148651	148652	Not provided	Not provided	66325.44884	5066.059886	3648.7672
148652	148653	Not provided	Not provided	66325.44884	5066.059886	3648.7672
148653	148654	Joe Lopez	Counselor, Log Cabin Ranch	0.00000	0.000000	-618.1300

148654 rows × 11 columns

Sekcija 5 Pandas - kombinovanje dataframe-a

Postoje tri glavna načina za kombinovanje dataframe-a : "Merging", "Joining" i "Concatenating".

Daćemo primere za svaki. Definišimo prvo tri nova dataframe-a:

In [75]:
```
df1 = pd.DataFrame({'A': ['A0', 'A1', 'A2', 'A3'],
                    'B': ['B0', 'B1', 'B2', 'B3'],
                    'C': ['C0', 'C1', 'C2', 'C3'],
                    'D': ['D0', 'D1', 'D2', 'D3']},
                   index=[0, 1, 2, 3])
```

In [76]:
```
df2 = pd.DataFrame({'A': ['A4', 'A5', 'A6', 'A7'],
                    'B': ['B4', 'B5', 'B6', 'B7'],
                    'C': ['C4', 'C5', 'C6', 'C7'],
                    'D': ['D4', 'D5', 'D6', 'D7']},
                     index=[4, 5, 6, 7])
```

In [77]:
```
df3 = pd.DataFrame({'A': ['A8', 'A9', 'A10', 'A11'],
                    'B': ['B8', 'B9', 'B10', 'B11'],
                    'C': ['C8', 'C9', 'C10', 'C11'],
                    'D': ['D8', 'D9', 'D10', 'D11']},
                   index=[8, 9, 10, 11])
```

In [78]:
```
df1
```

Out[78]:

	A	B	C	D
0	A0	B0	C0	D0
1	A1	B1	C1	D1
2	A2	B2	C2	D2
3	A3	B3	C3	D3

In [79]:
```
df2
```

Out[79]:

	A	B	C	D
4	A4	B4	C4	D4
5	A5	B5	C5	D5
6	A6	B6	C6	D6
7	A7	B7	C7	D7

In [80]:
```
df3
```

Out[80]:

	A	B	C	D
8	A8	B8	C8	D8
9	A9	B9	C9	D9
10	A10	B10	C10	D10
11	A11	B11	C11	D11

"Concatenation" praktično lepi dva dataframe-a jedan za drugi. U vidu moramo imati dimenzije dataframe-a po kojima vršimo spajanje ili "konkatenaciju".

In [81]:
```
pd.concat([df1,df2,df3])
```

Out[81]:

	A	B	C	D
0	A0	B0	C0	D0
1	A1	B1	C1	D1
2	A2	B2	C2	D2
3	A3	B3	C3	D3
4	A4	B4	C4	D4
5	A5	B5	C5	D5
6	A6	B6	C6	D6
7	A7	B7	C7	D7
8	A8	B8	C8	D8
9	A9	B9	C9	D9
10	A10	B10	C10	D10
11	A11	B11	C11	D11

In [82]:
```
pd.concat([df1,df2,df3],axis=1)
```

Out[82]:

	A	B	C	D	A	B	C	D	A	B	C	D
0	A0	B0	C0	D0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	A1	B1	C1	D1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	A2	B2	C2	D2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	A3	B3	C3	D3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	A4	B4	C4	D4	NaN	NaN	NaN	NaN
5	NaN	NaN	NaN	NaN	A5	B5	C5	D5	NaN	NaN	NaN	NaN
6	NaN	NaN	NaN	NaN	A6	B6	C6	D6	NaN	NaN	NaN	NaN
7	NaN	NaN	NaN	NaN	A7	B7	C7	D7	NaN	NaN	NaN	NaN
8	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	A8	B8	C8	D8
9	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	A9	B9	C9	D9
10	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	A10	B10	C10	D10
11	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	A11	B11	C11	D11

Merge radi sličnom logikom kao i "merging" dve SQL tabele. Na primer:

In [83]:
```
left = pd.DataFrame({'key': ['K0', 'K1', 'K2', 'K3'],
                     'A': ['A0', 'A1', 'A2', 'A3'],
                     'B': ['B0', 'B1', 'B2', 'B3']})

right = pd.DataFrame({'key': ['K0', 'K1', 'K2', 'K3'],
                          'C': ['C0', 'C1', 'C2', 'C3'],
                          'D': ['D0', 'D1', 'D2', 'D3']})
```

In [85]:
```
left
```

Out[85]:

	key	A	B
0	K0	A0	B0
1	K1	A1	B1
2	K2	A2	B2
3	K3	A3	B3

In [86]:
```
right
```

Out[86]:

	key	C	D
0	K0	C0	D0
1	K1	C1	D1
2	K2	C2	D2
3	K3	C3	D3

In [84]:
```
pd.merge(left,right,how='inner',on='key')
```

Out[84]:

	key	A	B	C	D
0	K0	A0	B0	C0	D0
1	K1	A1	B1	C1	D1
2	K2	A2	B2	C2	D2
3	K3	A3	B3	C3	D3

"Joining" je prigodan način za spajanje kolona dve tabele koje su potencijalno drugačije indeksirane u jedan dataframe. Dajmo primer za "joining":

In [87]:
```
left = pd.DataFrame({'A': ['A0', 'A1', 'A2'],
                     'B': ['B0', 'B1', 'B2']},
                      index=['K0', 'K1', 'K2'])

right = pd.DataFrame({'C': ['C0', 'C2', 'C3'],
                    'D': ['D0', 'D2', 'D3']},
                      index=['K0', 'K2', 'K3'])
```

In [88]:
```
left.join(right)
```

Out[88]:

	A	B	C	D
K0	A0	B0	C0	D0
K1	A1	B1	NaN	NaN
K2	A2	B2	C2	D2

In [89]:
```
left.join(right, how='outer')
```

Out[89]:

	A	B	C	D
K0	A0	B0	C0	D0
K1	A1	B1	NaN	NaN
K2	A2	B2	C2	D2
K3	NaN	NaN	C3	D3

Pandas Vežbanje:

- Učitati Salaries.csv.
- Ispitati strukturu Salaries.csv.
- Štampati deskriptivne statistike i analizirati ih.
- Obraditi nedostajuće vrednosti - menjati numeričke podatke medijanom a kategoričke sa "NO DATA".
- Koja je prosečna osnovna plata? (BasePay).
- Na kojoj poziciji u firmi je čovek sa imenom JOSEPH DRISCOLL?
- Koliko on zarađuje, uključujući i beneficije?
- Kako je ime najplaćenijoj osobi?
- Koja je prosečna osnovna plata (Base Pay) po godini?
- Koliko ima različitih poslovnih pozicija (titula)?
- Kojih 5 poslova je najustaljenije?